Luckily there's some sort of building up ahead. Sadly, in our country not many buildings have basements, but even ground level shelter is better than no shelter. Black rain is about to fall.

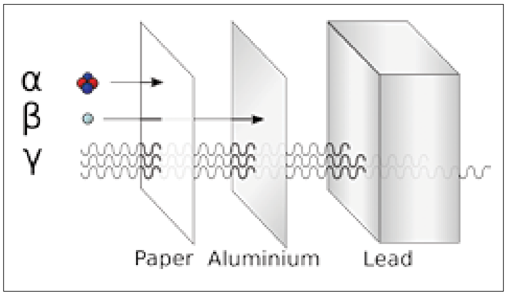
THE PROBLEM OF RADIATION AND LD_{50}

Ionizing radiation is bad because it disrupts our DNA. The sole physiochemical purpose of all the core molecules of life is to copy themselves before they die so that the metastructure can survive, and radiation messes with this replication. To save ourselves from it, we should know, what is this dreadful radiation and where does it come from?

Ionizing radiation is simply an emission of high enough energy that can knock out an electron or otherwise change the configuration of an atom with which it collides, thus hampering the molecule that it is a part of. It is given out during the nuclear fission blast along with the fission products (blast radiation), and later by spontaneous decay of the neutron-heavy daughter elements (residual radiation).

Ionizing radiation produced in nature is mainly of three types - alpha, beta and gamma. Alpha and beta radiation can be stopped by clothing, but gamma radiation needs a lot more than that! The blocking effect of materials on gamma radiation is quantified in the halving-thickness, i.e- the thickness that would absorb 50% of the radiation. Any shelter should have at least 10 layers of halving thickness to be effective. For example, packed earth has a halving thickness of ~10cm for gamma rays, so hiding a meter under the ground will reduce the intensity of the radiation 1024 times.

Radiation exposure is measured in Grays (Gy). 1 Gy is defined as radiation energy of one Joule absorbed per kilogram of mass. LD50 is the lethal dosage amount that would be fatal to 50% of the population. For the average healthy human LD50 is estimated at 3.5 Gy, and death is pretty much certain after exposure to doses higher than 6-7 Gy. Besides the radiation from the initial blast at ground zero, the danger of radiation is from neutron heavy fission products delivered over time by the fallout. A thermonuclear bomb vaporizes a lot of things in the explosion, resulting in the large cloud of dust and particulate matter that floats up in the hot air, to which radioactive molecules attach. The gradual deposition of this radioactivity (piggybacking on dust) over a large area is called fallout. In the first few weeks following the nuclear hit, fallout can result in a large area being uninhabitable. However, 70% of the millions of vaporised people were water, and combined with other sources of dihydrogen oxide, can lead to heavy enough droplets forming to cause rain locally within a few hours of the blast. This radioactive soot rich rain is what is called black rain.



Ionizing radiation

FALLOUT: NOT THE GAME

There's nothing like surviving a nuclear blast to put the essentials of life into perspective. Good old hunker down in the foxhole kinda stuff. Alpha and beta rays may not be able to get to you from the outside, but they can wreak havoc from inside your lungs or gut, so avoid the dust as much as possible. There might also be the stray lunatic so steel yourself for some serious human-hurting if needed. As you try to hide or get away from the black rain here's what you need to think about:

Shelter - This should be your top most priority. The ideal shelter is an underground bunker built and stocked for this very purpose, but most likely, you'll have to think on your feet. A building like the one you are in right now will shield you for the moment, but soon there will be decaying dust spewing radioactivity in every direction. If war had broken out some time before the nuclear strike, there may be blast shelters dug deep into the ground where you can camp. If buildings are all you have to hide in, find the largest one possible and go as far in as possible. The idea is to put as many walls (or as much material) as possible between you and the radioactive particles settling outside. While ventilation is normally a good thing, a room without any drafts to bring in the radioactive dust would be better. If the building is damaged, cover up holes with anything you can find lying around like cupboards, sheets of tin, etc. Also, choose a location that will not fill up with rainwater. If you have a few friends and a working crane, knock out one of the dispensers at a petrol pump after emptying it (which rioters may have already done for you) and hide in the storage tank buried underneath.

Another important factor, once you find a suitable shelter, is how long you have to stay there. While radioactivity does decrease with time, it's best to stay shielded for the first 48 hours minimum and do not venture out unless absolutely unavoidable. The area affected by fallout is greatly affected by weather and wind conditions, so you can start to leave the shelter within the week if you are upwind. If you are downwind, prepare to be holed in for a couple of weeks at the least!

Water and Food - You can survive three weeks without food but only three days without water. So you may have to leave the shelter to find potable water if you've run out. Urine is safe to drink upto three times before it becomes toxic, in case you get really thirsty before the first 48 hours are up. Even after the first 2-3 days, don't expose yourself to radiation for more than 30 minutes unless you're being evacuated. Under no circumstance should you ingest anything that is exposed to air. Anything in an airtight container or package is good to eat, so head to wherever the nearest grocery shop or supermarket was. Fruits with thick skins are okay too. Just make sure to get any dust off before hauling it out, and open it once you get back to your shelter. Be prepared to keep your scarce ration out of the hands of rats and other scavengers.

Comms and Apparatus - The EMP from a nuclear blast is more than likely to fry all solid state electronics con-